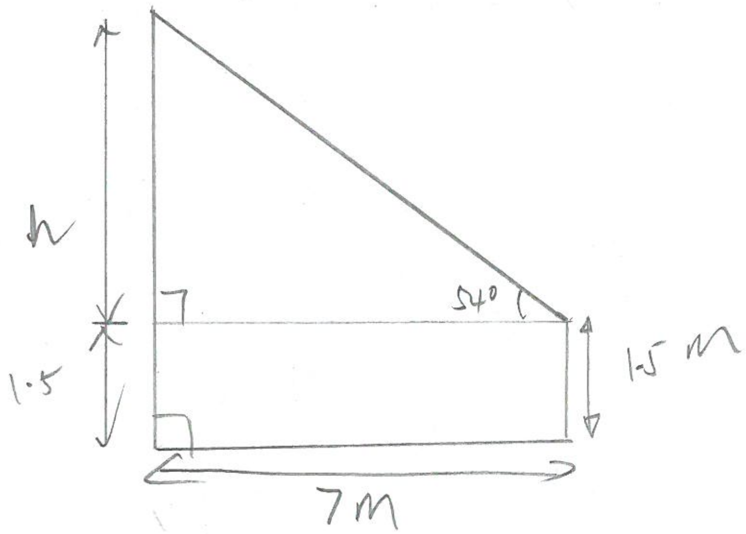
Revision – Trigonometry

1. Matthew, 1.5m tall, is measuring the height of his house. He is standing 7m from the base of the house and the angle of elevation is 54°.
a) Draw a labelled diagram to represent this situation.
b) Determine the height of the house to the nearest cm.

a)



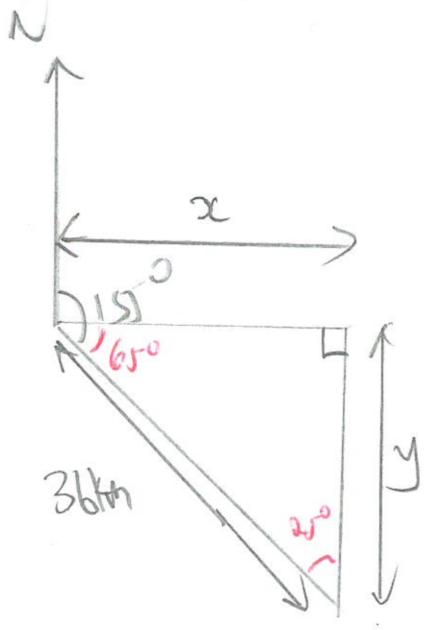
b) $\tan 54° = \frac{h}{7}$

$h = 7 \tan 54°$

$= 9.6346$ m

The height of the house is 11.13 m [1113 cm]

2. A boat travels for 36km in a direction of 155°T.
a) How far south does the boat travel?
b) How far east does the boat travel?



a) $\sin 65° = \frac{y}{36}$

$y = 36 \sin 65°$

$= 32.6$

The boat travels 32.6 km south

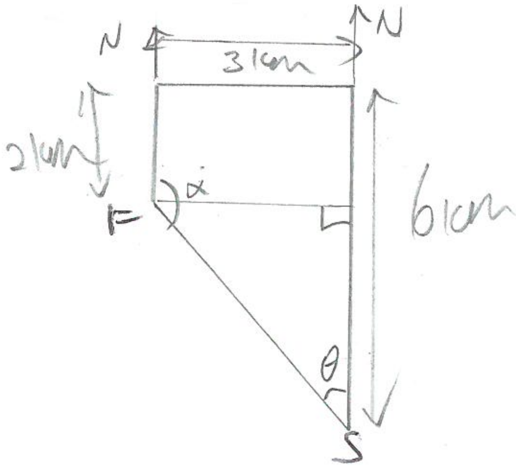
b) $\cos 65° = \frac{y}{36}$

$y = 36 \cos 65°$

$= 15.2$

The boat travels 15.2km east.

3. Mark cycles north for 6km, west for 3km and then south for 2km. What is his true bearing from his starting position?



$$\tan\theta = \frac{3}{(6-2)} = \frac{3}{4}$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right)$$

$$= 36.9^\circ$$

Mark's true bearing from his starting position $= 323.1^\circ T$